

Mel-Spectrograms and Data Augmentation for Spoken Digit Classification

Mohamad Melad Ali ASHAMES ¹[0000-0002-2837-1343] and Semih ERGİN ²[0000-0002-7470-8488]

^{1,2} Eskişehir Osmangazi University, Electrical, and Electronics Engineering Department,

Meselik Campus, 26480, Eskişehir/Turkey

¹muhashames@gmail.com,

²sergin@ogu.edu.tr

Abstract

A spectrogram is a visual representation of a time-domain signal's strength or "loudness" at various frequencies. It's an algorithm commonly used in audio spectral analysis and other fields. In this study, spectrograms were obtained and then mapped onto the Mel-Scale for the task of spoken digit analysis and classification. Gaussian noise was added to the audio data for the sake of data augmentation. Features were extracted from Mel-Spectrograms using a 7-layer Convolutional Neural Network (CNN), while the classification of these features was realized using two different classifiers: fully connected neural networks and Support Vector Machines (SVMs).

Keywords. Spectrogram, Mel-Scale, spoken digit, speech recognition, CNN, SVM.

1. Introduction

Speech recognition is a capability that enables a program to process human speech into a written or visualized format. In this study, we focused on the problem of English spoken digits identification. Nowadays spoken digit identification is an important task since it is used in various fields such as banking, social security, customer services, etc. In such fields we find ourselves obligated to list out the digits one by one over the phone to ensure that second parties can successfully access our information. English spoken digit recognition was performed in several studies. Bazzi and Katabi (2000) investigated the use of SVMs for English spoken digit recognition. Feature vectors were extracted from the Mel-Frequency cepstral coefficients of the speech signal and then reduced using principal component analysis (PCA). Becker et al. (2018) explored the efficiency of layer-wise relevance propagation (LRP) to identify relevant features for two neural network architectures that process either waveform or spectrogram representations of English spoken digits datasets. The results confirm that the networks are highly reliant on features marked as relevant by LRP. In a later study, Sharan used wavelet transform in forming the time-frequency representation (Sharan, 2020). The time-frequency representation is resized using bicubic interpolation to obtain scalograms which were used for recognizing spoken digits using CNN. The next sections of this paper will handle the use of Mel-Spectrograms, CNN, and SVM for English spoken digit processing and classification.

2. Materials and Method

There are many ways to represent audio data, like, waveform, Chroma features, Mel Frequency Cepstral Coefficients (MFCCs), Spectrograms, Mel-Spectrograms, and many more. In this study, the Mel-Spectrograms are used. Spectrograms that are scaled to the Mel-Scale have a closer representation of the human audio perception than other representations on the frequency scale. Thus, the Mel Spectrogram images of the spoken digits are extracted to be used as input to the CNN model.

2.1. Data

In this study, the Speech Commands dataset was used. It is a dataset for limited-vocabulary speech recognition by Google Brain (Warden, 2018). The data consists of audio files of spoken digits (from zero to nine)

and some other spoken words that were not used. Each digit has 2352-2377 audio files (Wav format). The length of each audio file is 1 second. Raw data were trimmed so that every digit has 2352 audio files each. A total of 23520 files were used in this study. Also, white (Gaussian) noise was added to each file which doubled the number of audio files (see section 2.4).

2.2. Spectrogram

The spectrogram is an important representation of audio data because human hearing is based on a kind of real-time spectrogram encoded by the cochlea of the inner ear (O'Shaughnessy, 1987). The spectrogram can be defined as an intensity graph (usually on a log scale, such as dB) of the Short-Time Fourier Transform (STFT) magnitude. Spectrograms are two-dimensional graphs, with a third dimension represented by colors. Time runs from left (past) to right (present) along the horizontal axis. The vertical axis represents frequency, which can also be thought of as pitch or tone, with the lowest frequencies at the bottom and the highest frequencies at the top. As it can be seen in Figure 1, the amplitude (or loudness) of a particular frequency at a particular time is represented by the third dimension, which is color, with dark blue corresponding to low amplitudes and brighter colors up through red corresponding to progressively stronger (or louder) amplitudes.

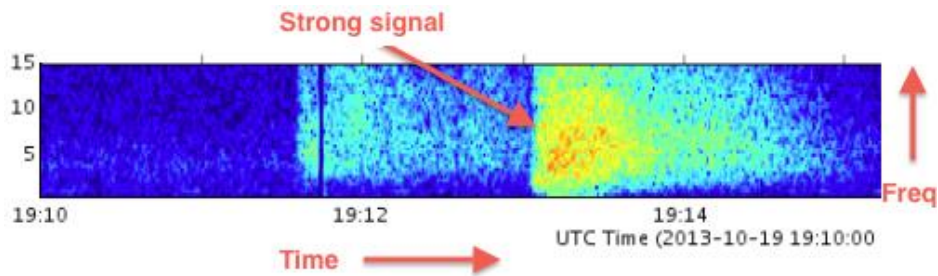


Fig. 1 Spectrogram dimensions: frequency, time, and color (Anonymous, 2021).

Sound is an analog signal, to obtain the spectrograms of sound signals they must be digitalized. Hence, each signal was sampled with a rate of 22050 samples per second. Figure 2 presents a waveform of a spoken digit belonging to the class (seven) where all the samples are shown, while Figure 3 only shows the samples between 0 and 100.

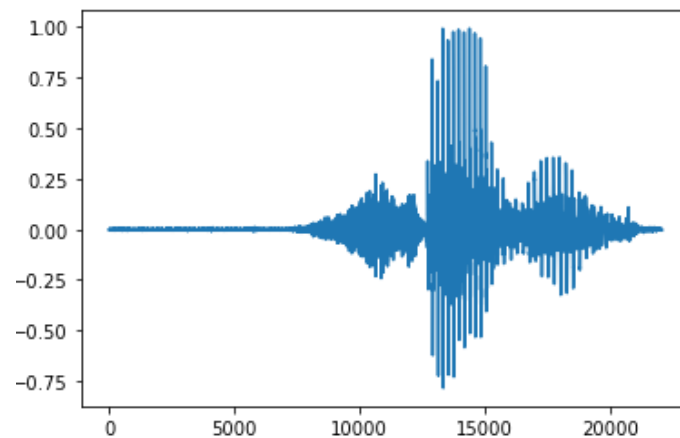


Fig. 2 waveform of a spoken digit belonging to the class (seven) samples with a rate of 22050 samples/second.

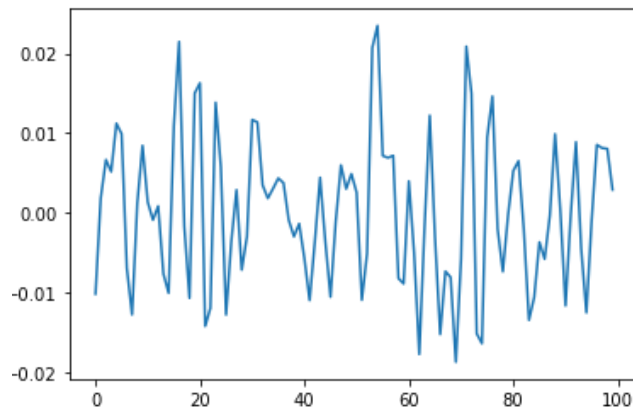


Fig. 3 A segment of 100 samples of the waveform in figure 2.

To generate a spectrogram, a time-domain signal is divided into shorter segments of equal length. Then, the fast Fourier transform (FFT) is applied to each segment (This is called Short-Time Fourier Transform). The term fast Fourier transform (FFT) refers to an efficient implementation of the discrete Fourier transform (DFT) for highly composite transform lengths. The STFT is simply a sequence of FFTs of windowed signal segments, where the windows are usually allowed to overlap in time, typically by 25-50%. The window length, as can be seen in Figure 4, determines the number of FFTs used to create the spectrogram. Plotting the spectrum of this sequence of stacked FFT segments creates the spectrogram.

To obtain the spectrograms, each spoken digit audio signal was divided into short time windows (segments) of 2048 samples (which corresponds to a physical duration of 93 milliseconds at a sample rate of 22050 Hz), making hops of size 512 samples each time to sample the next window, this way a %25 overlap is achieved.

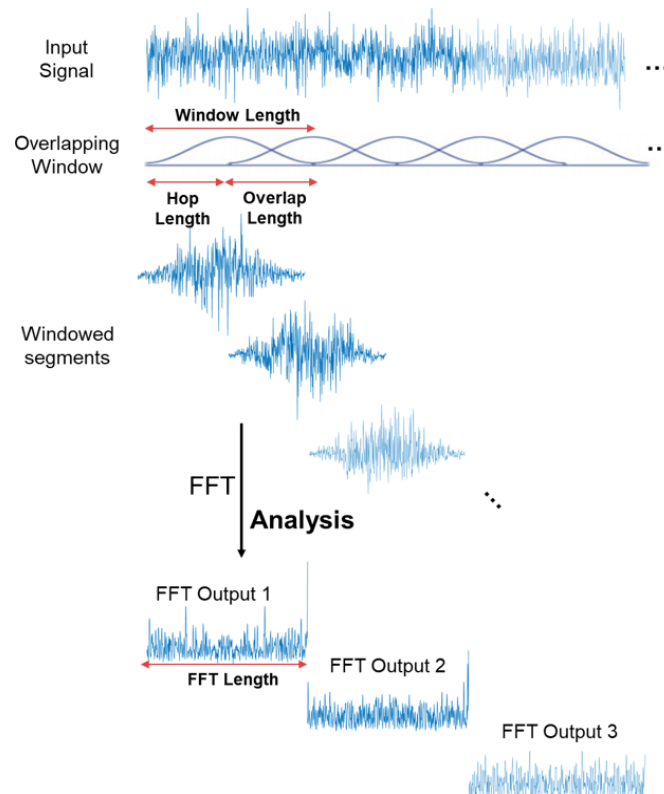
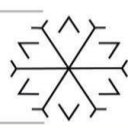
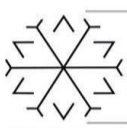


Fig. 4 Applying FFT on several windows of a particular number of samples (Anonymous, 2019).



Then, the FFT was applied to each window. Figure 5 shows a window of 2048 samples from the same waveform in figure 2.

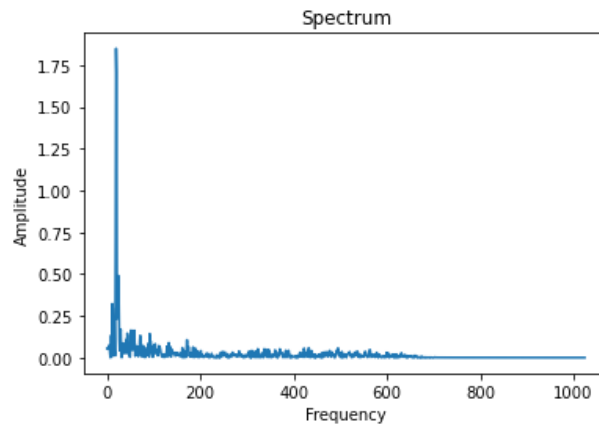


Fig. 5 A window of 2048 samples

Plotting the segments, we obtain the image in Figure 6(a). Here the y-axis is in Hertz (Hz). To obtain a spectrogram, we transform the y-axis (frequency) to log scale, and the “color” axis (amplitude) to decibels, as can be seen in Figure 6(b) where the spectrogram of the same signal above is illustrated.

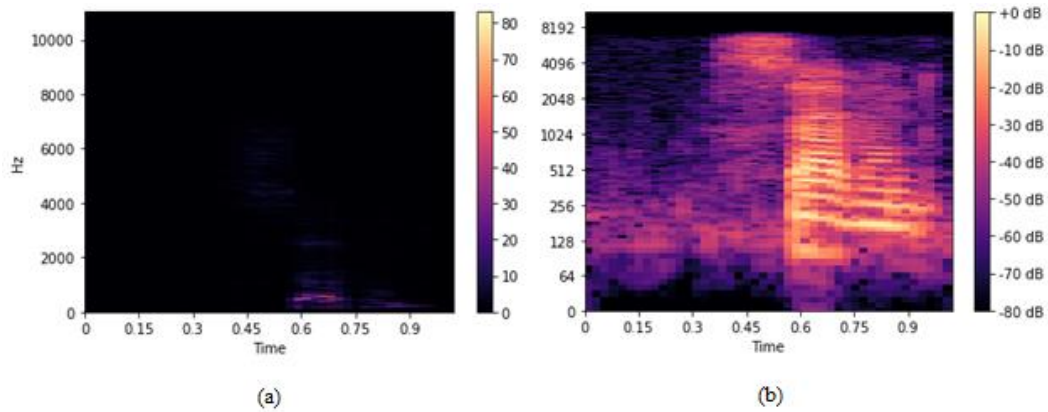


Fig. 6 (a) Representation of the matrix obtained from the FFT stack, (b) spectrogram of the same signal.

2.3. The Mel-Spectrogram

Humans are better at detecting differences in lower frequencies than higher frequencies. In more detail, a human can easily tell the difference between 500 and 1000 Hz but is hardly able to tell a difference between 10,000 and 10,500 Hz, even though the distance between the two pairs is the same. The Mel-Scale mimics humans' perception of sound. Sounds of equal distance on the Mel-Scale are perceived to be of equal distance to humans. The transformation from the Hertz scale to the Mel Scale is computed by Eqs. (1).

$$M = 1127 \times \log(1 + f/700) \quad (1)$$

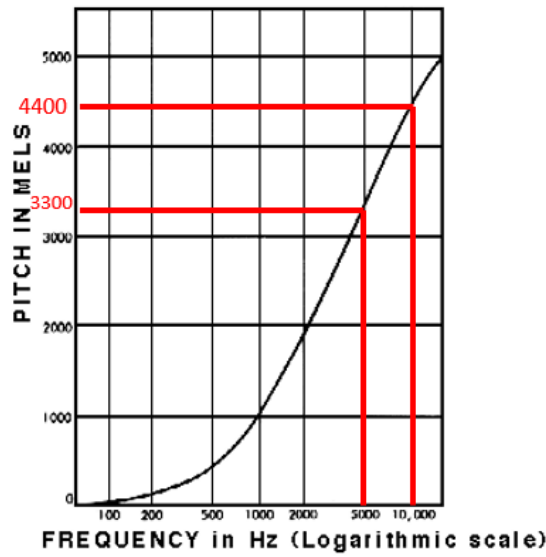


Fig. 7 The relationship between Hertz and Mels (Perera and Luening, 1975).

A Mel Spectrogram is a spectrogram where the frequencies are converted to the Mel-Scale (a Spectrogram with the Mel-Scale as its y-axis). Figure 7. Shows the relationship between Hertz and Mels. To generate a Mel-Scale, we separated the frequency spectrum into 128 evenly spaced frequencies (not by distance on the frequency dimension, but distance as it is heard by the human ear). Then we mapped the y-axis (frequency) of the obtained spectrograms onto the Mel-Scale to form the Mel-Spectrograms. A result Mel-Spectrogram of the same signal processed above can be seen in Figure 8.

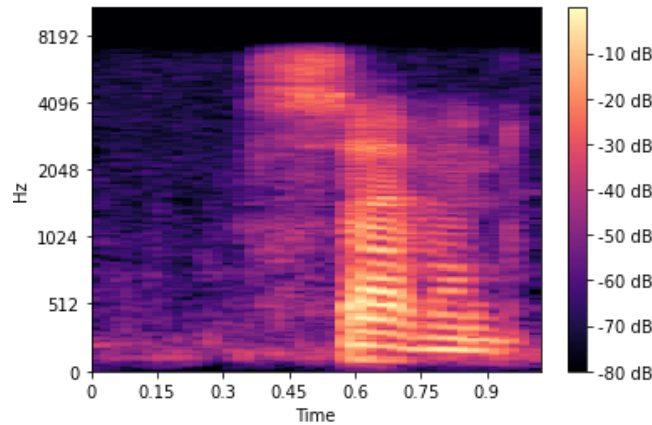


Fig. 8 Spectrogram of an audio file belonging to the class (seven)

2.4. Audio Augmentation

Since a 7-layer CNN model is used for extracting features from the obtained Mel-Spectrograms, the more input data the better. This helps with generalizing the model and preventing overfitting by providing more diverse features. To augment the audio data, Gaussian (white) noise was added to each spoken digit's audio file, resulting in a total of 47040 audio files when added to the original data (4704 for each class). Then, the Mel-Spectrogram of each noisy spoken digit audio file is created and added to the dataset of noiseless Mel-Spectrograms. Figure 9 (a) shows a Mel-Spectrogram of a spoken digit of the class (seven), and a Mel-Spectrogram belonging to its noisy version (b).

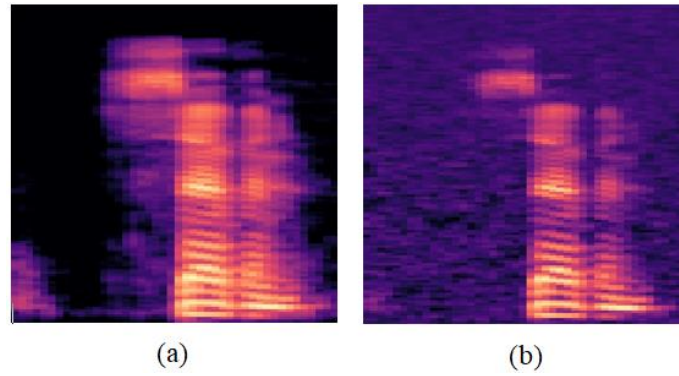


Fig. 9 Mel-Spectrogram of a spoken digit of the class “seven” (a), Mel-Spectrogram belonging to its noisy version (b).

2.5. Feature Extraction and Classification

To extract features from the Mel-Spectrograms, a 7-layer CNN architecture is designed. The input layer's size is $377 \times 387 \times 3$, which is the size of the created Mel-Spectrograms. The architecture consists of 6 convolutional layers with 3×3 kernels. The padding parameter is fixed to “1” at each convolutional layer. Every convolutional layer is followed by a batch normalization layer, a ReLU activation function, and a max-pooling layer except for the last convolutional layer which is followed by only a batch normalization and a ReLU layer. Max pooling layers have kernels of 2×2 size and a stride value of 2. The seventh layer is a fully connected layer for classification followed by a SoftMax function. The fully connected layer is then replaced with an SVM classifier, and the classification accuracy of both classifiers is compared.

2.6. Training Procedure

The dataset is partitioned into training, test, and validation sets by a rate of 0.7, 0.2, and 0.1, respectively. The training parameters in table 1 were held the same through the three experiments discussed in section 3.

Table 1. Training parameters

Training Parameters & Results	Value
Training Set Size	70%
Validation Set Size	10%
Test Set Size	20%
Mini Patch Size	10
Optimizer	SGDM
Hardware	6 GB GPU NVIDIA 1660TI
Learning Rate	Constant 0.0001
Number of Epoch	20

3. Results

Three different experiments were carried out. In the first experiment, only Mel-Spectrograms of the original dataset were used as input data, while in the second experiment only Mel-Spectrograms of noisy data were used. In the third experiment, an augmented dataset containing Mel-Spectrograms of both original and noisy data was used as the input of the CNN model. The validation accuracy and elapsed time of the training procedure belonging to each experiment can be seen in table 2. The classification results of both fully connected and SVM classifiers can also be seen in Tables 3 and 4, respectively.

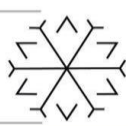
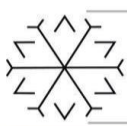


Table 2. Training Results.

Parameter	Original Data	Noisy Data	Augmented Data
Validation Accuracy	94.51%	92.2%	95.11%
Elapsed Time	34 min & 14 sec	34 min & 45 sec	69 min & 43 sec

Table 3. Test Results using 7-layer CNN.

Metric	Original Data	Noisy Data	Augmented Data
Accuracy (ACC)	94.6%	91.8%	95.3%
Specificity (SP)	100%	100%	99.6%
Sensitivity (SN)	99.3%	99.5%	99.5%
Positive Prediction Value (PPV)	1	1	0.996
Negative Prediction Value (NPV)	0.993	0.995	0.995
False Positive Rate (FPR)	0	0	0.003
False Negative Rate (FNR)	0.006	0.004	0.004
False Discovery Rate (FDR)	0	0	0.003
False Omission Rate (FOR)	0.006	0.004	0.004

Table 4. Test Results using 6-layer CNN model with an SVM classifier.

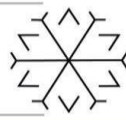
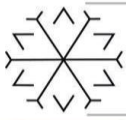
Metric	Original Data	Noisy Data	Augmented Data
Accuracy (ACC)	93.5%	90.3%	94.3%
Specificity (SP)	98.9%	99.4%	99.2%
Sensitivity (SN)	98.1%	98.7%	99.4%

4. Conclusion

The proposed spoken digit classification approach had been proven to be as successful as leading studies in the literature with an accuracy of 95.3% with the fully connected network as a classifier. The reached results show the efficiency of using Mel-Spectrogram features for speech recognition and classification. It was also proved that audio data augmentation help generalizes the CNN model and increase the overall classification accuracy. It is recommended to use the latest version of the used Speech Commands dataset which contains more data. In most cases, more input data boosts the classification accuracy and model generalizability. In addition, using Transfer Learning can also boost the reached results.

References

- Bazzi, I. and Katabi, D., 2000. Using support vector machines for spoken digit recognition. In Sixth International Conference on Spoken Language Processing.
- Becker, S., Ackermann, M., Lapuschkin, S., Müller, K.R. and Samek, W., 2018. Interpreting and explaining deep neural networks for classification of audio signals. arXiv preprint arXiv:1807.03418.
- Mathworks (2019) Short-time FFT. Available at: <https://www.mathworks.com/help/dsp/ref/dsp.stft.html> (Accessed: 01 December 2021).



O'Shaughnessy, D., 1987. Speech Communication, Human and Machine Addison Wesley. Reading MA.

Perera, R. and Luening, O., 1975. The development and practice of electronic music. Prentice Hall.

Sharan, R.V., 2020, December. Spoken digit recognition using wavelet scalogram and convolutional neural networks. In 2020 IEEE Recent Advances in Intelligent Computational Systems (RAICS) (pp. 101-105). IEEE.

The Pacific Northwest Seismic Network (PNSN) (2021) What is a Spectrogram?. Available at: <https://pnsn.org/spectrograms/what-is-a-spectrogram/> (Accessed: 01 December 2021).

Warden, P., 2018. Speech commands: A dataset for limited-vocabulary speech recognition. arXiv preprint arXiv:1804.03209.